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A massively parallel spatially resolved stochastic cluster dynamics method for simulations of irradiated materials ☆,☆☆

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ABSTRACT

The spatially resolved stochastic cluster dynamics (SRSCD) method is one of the most important methods for simulating the time-evolution of spatially correlated microstructures in irradiated materials. It is a kinetic Monte Carlo-based method for stochastically evolving integer-valued populations of defects within multi finite volume elements according to reaction rates. However, the increasing spatial scale and complexity of the simulated systems have exceeded the capabilities of serial SRSCD. To extend SRSCD to simulate large-scale complex systems, we propose a massively parallel SRSCD method in this paper and implement the program named MISA-SLSCD. It contains four contributions: (1) a dynamic Defect-Reaction Tree data structure to efficiently store and update millions of defect species and reactions; (2) a double-grouping search strategy to speed up the search for defects and reactions; (3) an adaptive synchronous algorithm to balance the accuracy and efficiency of simulations; and (4) an on-demand communication strategy to eliminate communication redundancy. A series of numerical simulation results show that our method has high accuracy in simulating damage accumulation in irradiated materials and obtains a good performance of over 90% parallel efficiency on 32,000 CPU cores with a 32 million-volume-element system.

Program summary
Program Title: MISA-SLSCD
CPC Library link to program files: https://doi.org/10.17632/ctmn5f5bzk.1
Code Ocean capsule: https://codeocean.com/capsule/8351487
Licensing provisions: BSD 3-clause
Programming language: Fortran90
Nature of problem: Behaviours of defects in irradiated materials are typically space-dependent long-term dynamical processes, and spanning multiple orders of magnitude in space (∼nm to m) and time (∼ μs to years). The SRSCD method is an important method for simulating the time-evolution of spatially correlated microstructures in irradiated materials. However, the increasing spatial scale and complexity of the simulated systems have exceeded the capabilities of serial and existing parallel versions. It is necessary to develop new massively parallel method and program to extend SRSCD to simulate large-scale complex systems.
Solution method: To extend SRSCD to simulate large-scale complex systems, we propose a massively parallel SRSCD method in this paper and implement the program named MISA-SLSCD. It contains: (1) a dynamic Defect-Reaction Tree data structure to efficiently store and update millions of defect species and reactions; (2) a double-grouping search strategy to speed up the search for defects and reactions; (3) an adaptive synchronous algorithm

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to balance the accuracy and efficiency of simulations; and (4) an on-demand communication strategy to eliminate communication redundancy.

1. Introduction

The performance failure (such as hardening, embrittlement, swelling) of nuclear materials mainly depends on the kinetic behaviour of radiation-induced defects, such as diffusion [1], nucleation, growth [2], and coarsening [3,4]. These behaviours are typically space-dependent long-term dynamical processes spanning multiple orders of magnitude in space (∼nm to m) and time (∼ μs to years) [5]. This typically exceeds the capabilities of atomic-scale simulation methods, such as molecular dynamics (MD) methods [6–8] (∼ps to ns, μm) and atomic kinetic Monte Carlo (AKMC) methods [9,10] (∼s to minutes, μm). It is necessary to use mesoscale methods to simulate the spatially resolved time-evolution of radiation defects. Cluster dynamics (CD) developed from mean field rate theory (MFRT) [11] is the commonly used mesoscale method. It is typically very computationally efficient and can model large doses (100 dpa or more) and timescales [12], and has a wide variety of applications in simulating microstructural long-term evolution [13–17]. However, the traditional CD loss of the spatial distribution of defects, and the number of rate equations to be solved increase exponentially with the number of species modelled.

Spatially resolved stochastic cluster dynamics (SRSCD) [18] has recently been developed as an effective alternate method for simulating the long-term evolution of inhomogeneous defects in irradiated materials. Based on the classical KMC algorithm, this method stochastically evolves integer-valued populations of defects within a mesh of finite volume elements according to reaction rates constructed from traditional CD. In SRSCD, defects are assumed to be homogeneously distributed within volume elements, which can dissociate, cluster, and be absorbed by sinks (such as grain boundaries and dislocations [19,20]). Mobile defects can also migrate between volume elements. This method was originally developed as a stochastic simulation algorithm (SSA) [21] to treat chemical master equations and has been widely used in many fields [22–24], and subsequently adapted for radiation defects in nuclear material by Marian et al. [25]. The spatial resolution of this method was developed by Dunn et al. [18] and shown to be effective in simulating the long-term evolution of radiation defects in spatially inhomogeneous systems [18,26–29].

However, mesoscale microstructure evolution is a very complicated process in irradiated materials. There are millions of species in irradiated materials whose behaviours span a much larger spatiotemporal scale, typically beyond the simulation scales of serial SRSCD. Therefore, a large gap exists between the simulation results of SRSCD and the actual physical phenomena. The simulation scales of SRSCD are generally limited by two factors: increasing spatial scale and complexity (defect types and interactions) of simulated systems. To reach larger simulation scales, parallel implementation of SRSCD is needed. However, the use of KMC causes asynchronism issues among different processes during parallel computation since time increments are randomly selected in various processes at each step. To solve this problem, Dunn et al. developed synchronous parallel SRSCD [30] (SP-SRSCD) by using the synchronous parallel (SP) KMC algorithm [31] to select the same time increment for all processes. But its communication load increases rapidly with the number of processes since synchronization and global communication are required at each step. To extend the capability of SRSCD to simulate large-scale complex systems and improve its computational efficiency, we develop a new massively parallel SRSCD method in this paper. In this method, we implement the parallelization of SRSCD based on the synchronous sublattice (SL) KMC algorithm developed by Shim and Amar [32]. The SL algorithm can synchronize after several KMC steps (with a time threshold), which theoretically requires only local communications and can achieve linear scaling; this algorithm has been widely used in parallel simulations of AKMC [33–36]. However, to the best of our knowledge, it hasn't been extended to parallelizing SRSCD. Our work in this paper is the first attempt to accomplish this.

In this paper, our parallel SRSCD method is applied to simulate radiation damage in Fe-based materials, such as α-Fe and Fe-Cu alloys. It can also be easily extended to simulations of other materials, such as the tungsten-based alloys used in fusion reactors. We aim to scale SRSCD to reach experimental spatiotemporal scales and radiation dose levels. However, there are several challenges for larger-scale parallel SRSCD simulations: (1) the number of defect species and reactions increase rapidly, even to millions, as the simulation space expands and time progresses [37]; (2) computational time is mainly spent on selecting reaction events; (3) it is difficult to select a proper time threshold for the SL algorithm to balance accuracy and parallel efficiency; and (4) communication overhead caused by data exchange and synchronization between processes affects scalability with increasing simulated space size and numbers of processes. We propose four optimization methods in our parallel SRSCD method to overcome the above challenging problems, and implement the program named MISA-SLSCD, including a dynamic Defect-Reaction Tree data structure, a double-grouping search strategy, an adaptive synchronous algorithm, and an on-demand communication strategy. With these methods, we successfully simulate the defect evolution in irradiated α-Fe and Fe-Cu alloys, obtain results that are in good agreement with those of other simulation methods and experiments, and achieve simulations of radiation dose levels equal to those approximately 40 years of operation of pressurized water reactors. The parallel efficiency of our method is significantly better than that of the SP-SRSCD [30], and the simulation scale is expanded to 32 million volume elements, which is 500 times that of SP-SRSCD. A strong scaling test shows that our parallel SRSCD method achieves over 92% parallel efficiency on 25,600 CPU cores, and a weak scaling test shows that the parallel efficiency exceeds 90% on 32,000 CPU cores.

The key contributions of this paper are as follows:

- We develop a massively parallel SRSCD method to extend SRSCD to simulate large-scale complex systems.
- We design a dynamic Defect-Reaction Tree data structure to efficiently manage massive number of defect species and reactions in SRSCD simulations.
- We develop a double-grouping search strategy to accelerate the selection of reaction events.
- We propose an adaptive synchronous algorithm to balance the accuracy and efficiency of parallel SRSCD.
- We propose an on-demand communication strategy to eliminate communication redundancy in large-scale simulations.

The paper is organized as follows. Section 2 introduces the theoretical background of SRSCD and discusses the related work. Section 3 discusses the new massively parallel SRSCD method and its optimization methods. The evaluation of accuracy and performance of our method is discussed through several numerical simulations in Section 4. Section 5 concludes our work.

2. Background

2.1. Spatially resolved stochastic cluster dynamics method

SRSCD is a promising method for simulating damage accumulation and diffusion in irradiated materials and approximately includes the spatial resolution of defects. In SRSCD, defect populations are evolved stochastically within a mesh of finite volume elements (VEs). Defect clusters are assumed to be homogeneously distributed within VEs [18] and evolve an integer-valued defect number within the volume. There are many potential reaction channels for these clusters, such as implantation, dissociation, annihilation at sinks, clustering, and diffusion between VEs. The relative reaction rates are derived from classical CD methods, and a classical KMC method (also known as the KMC direct method) is used to select the reaction event and time for the next step.

We briefly summarize the SRSCD method developed by Dunn et al. [18] for simulating radiation defect evolution. Consider a system containing N defect-clusters $\{S_1, \cdots, S_N\}$ that can participate in M reaction channels $\{R_1, \cdots, R_M\}$. Let $X(t)$ be the dynamic state vector of the number of potential defect species at an arbitrary time t, $X(t) = \{X_1(t), \cdots, X_N(t)\}$, where $X_i(t)$ is the number of defect clusters of type S_i at time t. Then, the evolution of the defect populations X takes the following form:

$$\frac{dX}{dt} = G + RX + KXX' + \nabla \cdot D\nabla X \tag{1}$$

where G is a source term, that represents the rates of 0^{th} (implantation) order reactions. R and K are matrices containing the reaction rates of 1^{st} (dissociation, annihilation at sinks) and 2^{nd} (clustering) order reactions, respectively. The last term represents the diffusion rate of mobile species, which is given by a finite-volume solution of Fick's law [26]. This equation can then be discretized in spaces using finite volume elements V, where each volume has its own vector of $X(V, t)$. Then, $\{G(V), R(V), K(V)\}$ become the rate matrices of the reaction channels in each volume V.

For clarity, the different types of reaction channels are briefly described as follows. Readers can refer to the original paper [18] for more details about the calculations of reaction rates.

- 0^{th}`-reaction`: $0 \to S_i$. Implantation of initial defects, such as Frenkel pair (vacancy and self-interstitial atom (SIA)) implantation (under electron irradiation) or displacement cascade (contains vacancies, SIAs, and their clusters) implantation (under neutron irradiation).
- 1^{st}`-reaction`: $S_i \to 0$ or $S_i \to S_1 + S_{i-1}$. Defects annihilate at sinks, or a defect cluster dissociates a monomer.
- 2^{nd}`-reaction`: $S_i + S_j \to S_k$. Clustering of two defects.
- `Diffusion`: $S_i(V_1) \to S_i(V_2)$. Defects diffuse from one VE to another. There are at most six directions of diffusion. In the context of modelling and simulation of irradiated materials, defect species S include mobile species S_m and immobile species S_{im}, and only mobile species can diffuse.

Instead of directly solving Eq. (1), the classical KMC algorithm is used to stochastically evolve X. In this algorithm, only one reaction event occurs in a time step, and the time increment Δt and selected reaction R_j are determined by two random numbers r_1 and r_2, $r_1, r_2 \in (0, 1]$, as the following formulas:

$$\Delta t = \frac{1}{a_{\text{tot}}(X)} \ln\left(\frac{1}{r_1}\right) \tag{2}$$

$$\sum_{\mu=1}^{j-1} a_\mu(X) < r_2 a_{\text{tot}}(X) \le \sum_{\mu=1}^{j} a_\mu(X) \tag{3}$$

where $a_{\text{tot}}(X)$ is the total reaction rate of a KMC domain and a_μ is the reaction rate of R_μ. Then, the system changes caused by reaction event R_j within the time interval $[t, t + \Delta t)$ can be expressed as:

$$X(t + \Delta t) = X(t) + v_j \tag{4}$$

where v_j is the state-change vector of R_j, which represents the +/− number of defects caused by reaction event R_j.

2.2. Related work

Since SRSCD relies on the classical KMC algorithm to stochastically evolve defect populations, the parallelism of SRSCD is the parallel implementation of the KMC algorithm. There has been much research on KMC parallelism, including parallel algorithms [31,32,38,39] and hardware-based (such as GPU, and SW26010 many-core processors) parallel implementations [33,40]. This paper mainly focuses on parallel algorithm aspects.

Several parallel KMC algorithms have been proposed to expand the simulation scale of KMC, including the SP algorithm [31], synchronous relaxation (SR) algorithm [38], optimistic synchronous relaxation (OSR) algorithm [39], and SL algorithm [32]. These algorithms are all based on spatial decomposition, and mainly solve two critical issues of parallel KMC: (1) conflicts between neighbouring subdomains caused by boundary events and (2) synchronization between subdomains. However, the SP algorithm requires synchronization and global communication at each KMC step, which affects the computing performance. The SR algorithm can synchronize after a time interval, but the overhead of handing domain interactions is very large [41]. Then, Michael and Kristen proposed this optimistic SR algorithm to optimize the handling of boundary events. However, the OSR algorithm requires $2 \sim 3$ global communications in each synchronization cycle [41]. Of these, the SL algorithm requires only local communication [3] after a time interval and thus can obtain significantly better scaling than the SR/OSR and SP algorithms. During the past decades, it has been widely used in AKMC systems [42,43,33–36], such as SPPARKS (a general framework for parallel simulation of KMC), Crystal-KMC [43], and OpenKMC [33]. We introduce it to parallelization of SRSCD for the first time.

Besides the KMC-based SRSCD, classic CD methods are widely used to simulate radiation damage in materials [13,15,44,45]. This is based on a mean-field assumption and tracks defect evolution via rate equations for the concentration of defect to simulate the evolution of spatially inhomogeneous defects. Although several spatially resolved CD methods have been proposed [25,46–48]. But they are implemented serially, so the simulation scale is limited. Xu et al. [47] developed an efficient CD program named PARASPACE using OpenMP to implement spatially resolved CD in parallel. However, it supports only one-dimensional spatial resolution. The ODE-based CD methods all face challenges in simulating complex defect species due to computational requirements. Different from classic CD methods, we realize the large-scale simulation of radiation damage in materials based on the SRSCD method. Moreover, with our designed structure and the proposed efficient algorithm and strategies, our parallel SRSCD method can be easily extended to the simulation of complex systems without a significant increase in computational requirements.

3. A massively parallel SRCSD method

To extend the simulation scales of SRSCD, we employ the SL algorithm developed by Shim and Amar [32] to scale SRSCD across massive processes on a distributed parallel computing platform. In SL, different parts of the simulation box are assigned via spatial decomposition to different MPI (message passing interface) processes. To avoid conflicts between processes due to the synchronous nature of the algorithm, each subdomain is further divided into different sectors or sublattices and numbered (see Fig. 1) (four in 2D or eight in 3D). A complete synchronization cycle corresponding to a time interval τ is then as follows.

(1) At the beginning of a cycle, each process's local time is initialized to 0.
(2) One of the sectors is randomly selected so that all processes operate on the same sector during that cycle.
(3) Each process simultaneously and independently carries out KMC events in the selected sector until the time of the next event ex-

ceeds τ. As in the serial KMC, each event is carried out with time increment $\Delta t_i = -\ln\left(r_i\right)/R_i^{tot}$, where r_i is the random number between 0 and 1, and R_i^{tot} is the total reaction rate in the elected sector i.

(4) Each process communicates with its neighbouring processes about the necessary changes caused by boundary events, updates its configuration, and moves on to the next cycle using a new randomly chosen sector.

(5) When all four (or eight) sectors are executed, the entire system time is advanced by τ.

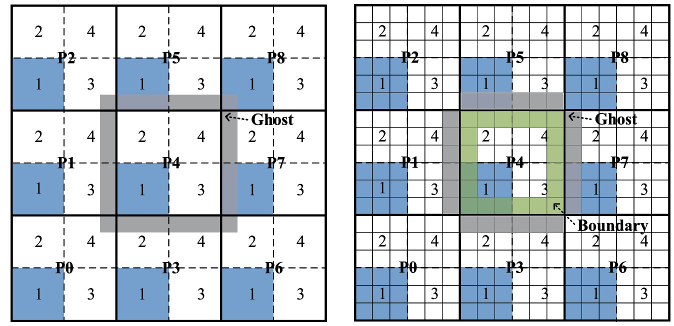
In the original SL algorithm, the sector is randomly selected at the beginning of a cycle. To ensure that each process selects the same sector, global communication is inevitably introduced. In our method, sectors are traversed sequentially starting from sector 1 for simplicity, just as in [35,36]. We avoid conflicts caused by boundary events by dividing sectors within subdomains and making the sector size larger than the length of the VE. It is obvious that the active sectors in neighbouring processes are always spatially separated by one sector, which avoids simultaneously modifying the same configuration in the same VE or VEs within the boundary region. While one selected sector proceeds, the other sectors are frozen during the cycle. Thus, boundary events that have occurred in neighbouring processes will not affect the execution of the current process.

As shown in Fig. 1, there is one more decomposition layer in our parallel SRSCD, which is used to solve the Fick term in SRSCD. After dividing subdomains and sectors, VEs are assigned to different processes and sectors via their central coordinates. This extra decomposition will not increase communication since in each cycle, the serial KMC proceeds as in the original SL algorithm. Only randomly select an event in one VE at a time step. In addition, since mobile defects can diffuse between VEs, the configuration in neighbouring processes must be known as far as the range of interaction for each process to calculate its boundary diffusion events. As a result, in addition to containing the configuration of its own domain, each process also contains a "ghost region" that includes the neighbouring process's relevant configuration beyond the process's boundary. At the end of each cycle, each processes communicates with its neighbouring processes and exchanges information to properly update its corresponding boundary and ghost regions. It can be seen that our ghost region is slightly different from that in the original SL algorithm. The original algorithm is proposed to simulate stochastic particle systems efficiently. Its ghost region usually surrounds the subdomain; thus, the process needs to communicate with its face-adjacent, edge-adjacent, and corner-adjacent neighbouring processes. In parallel SRSCD, communication is mainly caused by the diffusion of mobile defects, which is driven by concentration gradients of defects between face-adjacent VEs. Therefore, processes only need to communicate with face-adjacent neighbouring processes in our method, and the ghost region only contains face-adjacent VEs.

To improve the computational efficiency and scalability of our parallel SRSCD method, we develop several optimization methods, which provide an integrated solution, including a dynamic Defect-Reaction Tree data structure, a double-grouping search strategy, an adaptive synchronous algorithm, and an on-demand communication strategy. The following subsections discuss each of these in detail.

3.1. Defect-reaction tree data structure

As discussed previously, the defect populations need to be updated after each reaction event throughout simulations. Then, the possible reactions are updated, and the associated rates are recomputed. At the beginning of a simulation, the number of defect species and possible reaction may be few or none. However, in the context of irradiation, the number rapidly increases to thousands and even millions [37]. A well-designed data structure is required to efficiently store and quickly search through populations.



(a) spatial decomposition in original SL algorithm **(b) spatial decomposition in this parallel SRSCD method**

Fig. 1. Illustration of spatial decomposition (2D) in (a) the original SL algorithm and (b) our parallel SRSCD method. Heavy lines correspond to subdomains, dashed lines indicate sectors, and light lines correspond to VEs. Sectors coloured blue are selected in the current cycle. Grey and green represent ghost and boundary areas, respectively. Note that the 3×3 VEs in each sector are just an example. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

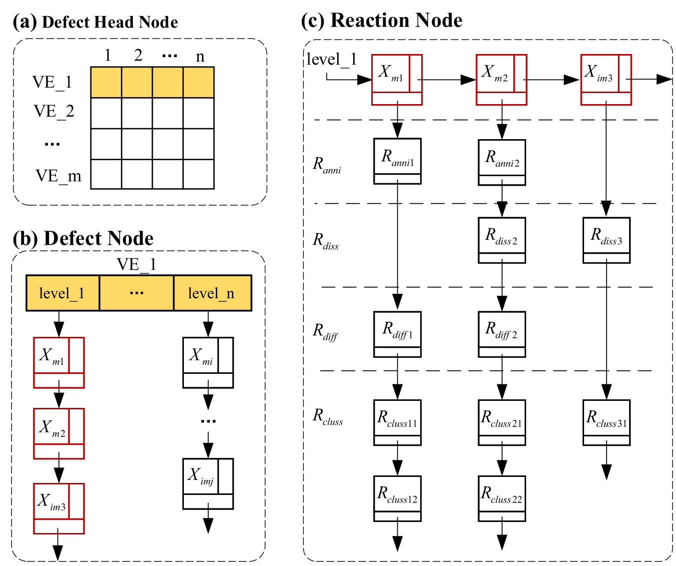


Fig. 2. Illustration of Defect-Reaction Tree data structure.

In fact, the possible reactions in the system are determined by the defect species, and a species mostly participates in four types of reactions (dissociation, annihilation at sinks, diffusion, and clustering). Only a small subset of defects and reactions are affected by a reaction event. Additionally, only one reaction event in one VE is executed in a step of SRSCD simulation.

Therefore, we can consider to associate a defect species in a VE with its reactions and index different VEs through an array to improve search efficiency. The defects can be further classified according to their components species. Based on these physical characteristics, we design a new data structure, called the **Defect-Reaction Tree (DRT)**, to achieve efficient storage and searching of defect species and reactions in SRSCD simulations. We classify species in a VE into n levels according to their type and store them with an array of linked lists. Each node in these lists represents a defect species. Then, each defect node maintains a reaction list, that contains the possible reactions taking the species as reactants. To reduce memory usage, we combine similar reactions and eliminate the storage of reactants of the same type. Fig. 2 illustrates the DRT structure, where X_m and X_{im} represent mobile and immovable species, respectively. Our DRT structure contains three dimensions: (1) defect

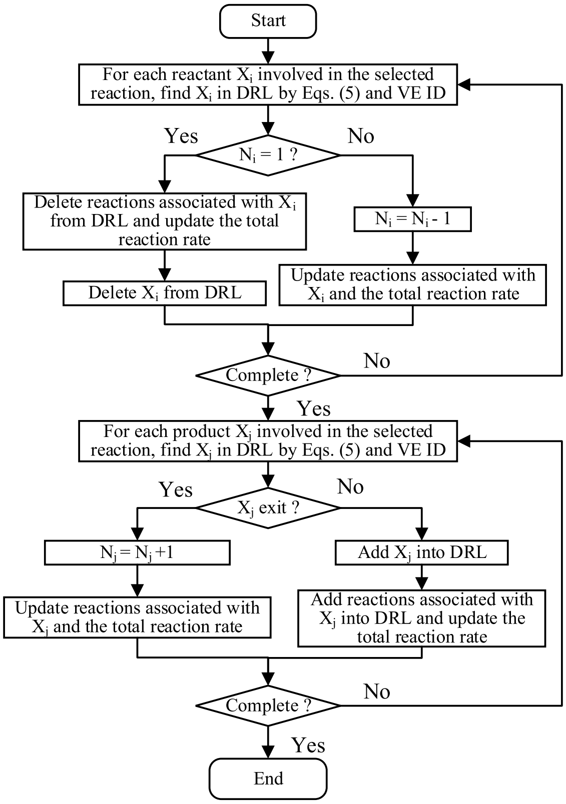


Fig. 3. Flowchart of DRT updating.

head node, (2) defect node, and (3) reaction node, which are described as follows. And the workflow for updating DRT is shown in Fig. 3.

- `Defect Head Node`: As shown in Fig. 2(a), the defect head node a two-dimensional array $L[m,n]$ that stores the header nodes of defect lists in each VE. Where m is the number of VEs in the current process, n is the number of levels given by $2(2^k-2)$, and k is the maximum number of component species in a defect cluster. Which level a defect species is located can be quickly determined by

$$s_i' = \begin{cases} 1, & s_i \neq 0 \\ 0, & s_i = 0 \end{cases}, i = \{1,2,\dots,k\}$$
$$\text{level} = \begin{cases} \sum_{i=1}^{k} s_i' 2^{k-i} + 2^k - 2, & \text{defect contain vacancies} \\ \sum_{i=1}^{k} s_i' 2^{k-i}, & \text{other defects} \end{cases} \tag{5}$$

where s_i is the number of component species i.
- `Defect Node`: As shown in Fig. 2(b), a defect node is a species, including information of defect type, number of defect-clusters of this type, diffusivity, etc. The defect type is a one-dimensional array of the number of component species that form the clusters. More specifically, the first digit represents the number of self-point defects in the cluster (positive value for SIAs and negative value for vacancies). For example, a cluster formed by one vacancy and three copper (Cu) atoms has the type $\{-1,3\}$, while $\{2,0\}$ represents a cluster with only two SIAs.
- `Reaction Node`: As shown in Fig. 2(c), all reaction types in which a species participates are linked to the defect node. Each reaction node contains the number of reactants and products, necessary type of reactants or products, and reaction rates. For memory-saving purposes, reactions of the same type are stored in one node. For example, a complex cluster V_iCu_j will dissociate a vacancy to form $V_{i-1}Cu_j$, or dissociate a Cu atom to form V_iCu_{j-1}. The two dissociation reactions have the same reactant but different products and reaction rates. Thus, the reactant needs to be stored only once in the defect node.

There are also some other data structures used to manage the above defect species and reactions. The developer of SCD used the UTHASH macros to construct two hash tables to manage all species and mobile species in the whole system respectively, and use a reaction matrix to store all reactions [25]. However, the entire reaction matrix needs to be reconstructed after each step, which leads to a high computational overhead since the number of defects and reactions in a simulation will increase to thousands and even millions. To overcome this problem, Hoang et al. [37] used a reaction hash table to replace the reaction matrix. However, each reaction needs to be visited to remove those whose reactants no longer exist due to the executed event, which also results in additional computational overhead. Hash tables can support O(1) operations, but the processing of hash conflicts and the calculation of hash values usually bring a lot of computational overhead. In addition, in nuclear material simulations, the number of defect species and reactions will increase first and then decrease, which will also bring overhead of hash expansion and low utilization of hash table space [49]. At the same time, hash information also requires additional memory. Considering the spatial resolution in SRSCD, extending the hash tables used in these SCD codes to the SRSCD requires considerable optimization to minimize the additional memory overhead. To manage data of defect species and reactions in SRSCD, Dunn et al. constructed two separate singly linked lists for defect species and reactions of each VE in their SP-SRSCD code [30]. However, this results in $\mathcal{O}(N)$ complexity for searching both species and reactions, and reactants are stored repeatedly in different reactions resulting in high memory usage.

While in our DRT structure, defect species are stored by levels in each VE. **The VE and level of a species can be located directly through the array index, with almost no computational overhead**. Then locating a species in a level may require $\mathcal{O}(N)$ complexity, but **the number of species in a level of a VE is usually only tens or even less (several), so the search for a species is very fast**. With the association of reactions with defect species, **the reactions of a species can be updated sequentially with $\mathcal{O}(1)$ complexity** without requiring frequent access to the entire data table. In addition, the information of each species and reaction type is well organized to reduce memory usage. Our DRT structure can not only be rapidly expanded to simulate systems with multiple defect species but also can be easily applied to simulations of other reaction-diffusion systems.

3.2. Double-grouping search strategy

As shown in Eq. (3), a reaction event is chosen in a step according to the total reaction rate of a KMC domain. In our parallel SRSCD method, a KMC domain corresponds to a sector. When m VEs with s species and n possible reactions per VE are present in a sector, $\mathcal{O}(mn)$ computation time is required to select a reaction event. In this work, all reactions in a KMC domain are first grouped according to the VE in which they are present, and the total reaction rate of each VE is recorded. We can first select a VE based on the total rate of each VE with $\mathcal{O}(m)$ computation time and select a reaction within the VE based on the rate of reactions with $\mathcal{O}(n)$ computation time. Then, the search time reduces from $\mathcal{O}(mn)$ to $\mathcal{O}(m+n)$.

Furthermore, reactions are connected to their reactant in the DRT structure. Thus, reactions in a VE can be further grouped by defect species, and the total rate of all possible reactions for a given species is recorded. This requires a computation time of only $\mathcal{O}(\frac{n}{s})$ to select a reaction in a VE. With the two groups, **the time of the selection reaction is significantly reduced, to $\mathcal{O}(m+\frac{n}{s})$**. We call these two grouping operations the **double-grouping search strategy**, which can bring considerably improve the computational efficiency for SRSCD simulations.

3.3. Adaptive synchronous algorithm

When using the SL algorithm, the time interval τ is a crucial factor affecting accuracy and parallel efficiency. If τ is small, synchronization occurs after a few reaction events in the sector, which results in high accuracy, but the parallel efficiency decreases. In contrast, if τ is increased, the synchronization frequency is reduced, and the parallel efficiency is greatly improved, but the accuracy is low. Since the reaction events lead to a change in reaction rates during the simulation, it is not easy to select a proper constant τ to balance the accuracy and parallel efficiency. Here, we introduce a self-correcting method for time interval τ.

First, the time step of a reaction event can be obtained through Eq. (2). Integrating Eq. (2) over (0,1], we have

$$\int_0^1 \Delta t dr = \int_0^1 \frac{1}{a_0} \ln\left(\frac{1}{r}\right) dr = \frac{1}{a_0} \tag{6}$$

Then, the number of events executed in the time interval τ is $n = \tau \times a_0$. a_0 is the total rate of all possible reactions in a sector. Ensure that a_0 of each sector is not much different, the load is balanced. Assume that the average number of reaction events executed in a cycle is n_{reac}. Let $\overline{a}_i$ be the average rate of each reaction event in sector i, where $i \in \{0, \cdots 7\}$. Define $a_{\max}$ as the maximum $\overline{a}_i$ of the processes, and $a_{\max all}$ as the maximum $a_{\max}$ across all processes. Then, the time interval τ can be given by

$$\tau = \min\left\{n_{reac}/a_{\max all}, T_{stop} - t\right\} \tag{7}$$

where T_{stop} is the total simulation time and t is the current simulation time.

Our **adaptive synchronous algorithm** based on the self-correcting τ is shown in Algorithm 1. With the dual factors of n_{reac} and reaction rates, the adaptive synchronous algorithm can obtain good accuracy and high parallel efficiency.

Algorithm 1: Adaptive Synchronous Algorithm.

Input: T_{stop} - total simulation time.
Input: t - the current simulation time.
Input: n_{reac} - proportion of the number of events executed.
while $t < T_{stop}$ **do**
  $a_{\max} \leftarrow 0$
  for $i = 0$ *to* 7 **do**
    $a_{0,i} \leftarrow$ total rate of possible reactions across sector i
    $n_i \leftarrow$ total number of reactions across sector i
    $\overline{a}_i \leftarrow a_{0,i}/n_i$
    $a_{\max} \leftarrow \max(\overline{a}_i, a_{\max})$
  $a_{\max all} \leftarrow$ maximum $a_{\max}$ across all processes
  if $a_{\max all} > 0$ **then**
    $\tau \leftarrow n_{reac}/a_{\max all}$
  else
    $\tau \leftarrow T_{stop} - t$
  $\tau \leftarrow \min(\tau, T_{stop} - t)$
  for $i = 0$ *to* 7 **do**
    $t_{kmc} \leftarrow 0$
    while $t_{kmc} < \tau$ **do**
      Compute Δt
      if $t_{kmc} + \Delta t > \tau$ **then**
        Exit
      else
        Choose a reaction in DRL
        Update DRL
        $t_{kmc} \leftarrow t_{kmc} + \Delta t$
    Communication and synchronization
  $t \leftarrow t + \tau$

3.4. On-demand communication strategy

In the simulation of irradiated materials with SRSCD, implantation under neutron irradiation is different from other reactions. For the former, a collection of cascade defects will be implanted into a VE and

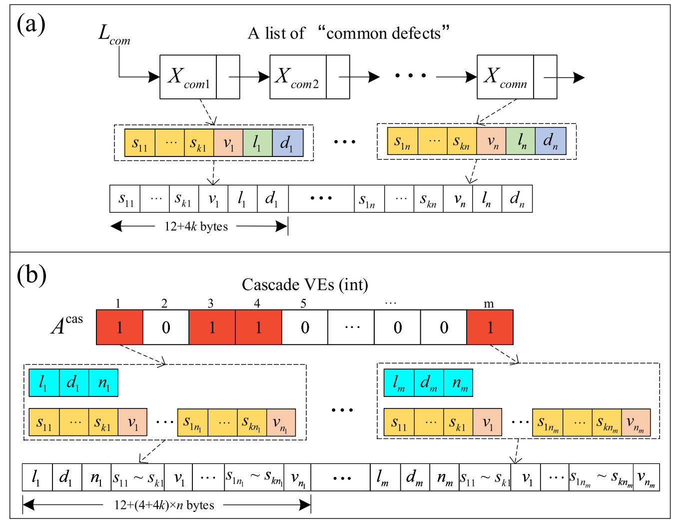


Fig. 4. Schematic diagram of data packaging for communicating (a) “common defects”, and (b) “cascade defects”.

immediately compounded with the existing defects, which results in an update of all defects in the VE. For the latter, the number and species of products can be known from the reactants. Only the species that participated in the selected reaction must be updated. We define the defects affected by the implantation event under neutron irradiation as “cascade defects”, denoted by $X_{cas} = \{X_{cas1}, X_{cas2}, \ldots, X_{casm}\}$, while the reactants and products of other reactions are defined as “common defects”, denoted by $X_{com} = \{X_{com1}, X_{com2}, \ldots, X_{comn}\}$. The following discussions in this subsection are all about neutron irradiation simulations.

In a synchronous cycle, if the reactants and/or products of selected reactions are in the ghost and/or boundary VEs (see Fig. 1(b)), they need to be recorded and sent to neighbour processes after the end of this cycle. The reactions associated with these changed defects are subsequently updated. Generally, only a tiny part of defect species in the ghost and boundary VEs are affected by the reaction events during a cycle. It should be noted that the defects in ghost VEs are affected only by diffusion from boundary VEs to ghost, and only mobile species participate in diffusion. Similarly, defects in the boundary VEs affect only diffusion from the neighbour processes to this process. If all defects in the boundary and ghost VEs are sent to neighbour processes, an increasing communication redundancy would be introduced as the spatial scale increases. **To eliminate communication redundancy, we propose an on-demand communication strategy. This strategy communicates only mobile species in ghost and boundary VEs that are affected by reaction events according to the simulation state.**

To correctly swap defects with neighbour processes, a temporary linked list L_{com} is created for recording X_{com} while using an array A_{cas} to record the boundary VEs that have X_{cas}. Each VE of L^{com} contains the associated information of the species to be updated. The steps for recording defects within the boundary and ghost VEs that need to be communicated are as follows:

(1) Set the initial value of A_{cas} to 0.
(2) Visit the selected reaction in the current cycle. If the selected reaction is implantation, go to (3); otherwise, go to (4).
(3) Visit each VE in L_{com}, and remove those whose VE ID equals the VE that occurs after implantation. Subsequently, set the value of A_{cas} to 1 with the VE ID as the index if the VE is located in a boundary area.
(4) Visit each reactant and product of the selected reaction, and check whether implantation has occurred in its VE in the previ-

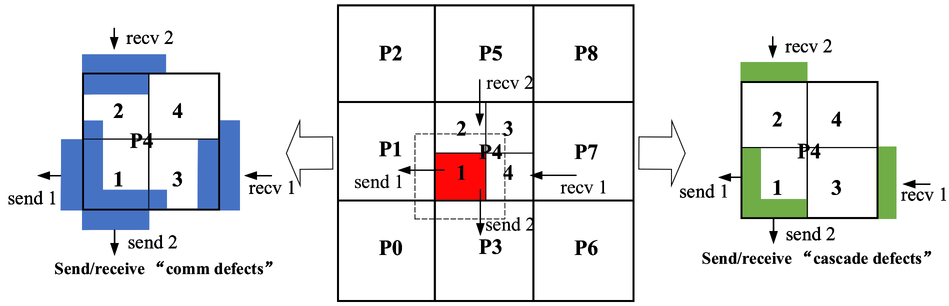


Fig. 5. Schematic diagram of 2D process communication between process P4 and its neighbours after sector 1 completes the calculation.

ous event(s). If it has, ignore it. Otherwise, add an element to L_{com} for the reactant or product.

Fig. 4 shows the reorganization of defect species to be communicated according to L_{com} and A_{cas}. Each data block L_{com} contains the type ($s_1 \sim s_k$) and changed number (v) of defect species, the local ID (l) of the current VE, and its neighbour's ID (d), as shown in Fig. 4(a). For X_{cas}, reorganization is more complex because the number of defect species and their population count in each VE are different. An additional index (see blue data block in Fig. 4(b)) is created for each VE in A_{cas} recorded as “1”. It contains the local IDs of the current VE and its neighbour's ID, and the total number of mobile defect types in the VE.

As discussed before, the diffusion of mobile defects between VEs is driven by the concentration gradient. Thus, communication occurs only between the surface-adjacent processes. As an example, Fig. 5 shows the communication after sector 1 completes its calculation. X_{com} and X_{cas} are swapped with two independent point-to-point communications. With the domain decomposition shown in Fig. 1(b), it needs only to send to three neighbour processes and receive from the three opposite neighbours during a synchronous cycle. Since the size of the messages to be sent is determined during running, the receiver does not know the information before receiving the messages. Thus, the *MPI_Probe* is used to query the information beforehand and then to determine whether to launch *MPI_Recv* to receive the messages.

Furthermore, if the received X_{com} belongs to the ghost VEs of a neighbour process, it does not belong to the neighbour who has already communicated. It is necessary to let the other neighbour processes know that their ghost VEs have changed through a second communication. This will increase the communication load, but a second communication is rarely needed.

4. Evaluation

In this section, a series of numerical simulations are conducted to evaluate the accuracy, memory usage, performance, and scalability of our parallel SRSCD method with the MISA-SLSCD program. The simulations are carried out on the SunRising-1 computing platform at the Computer Network Information Centre (CNIC), Chinese Academy of Sciences (CAS). Each node has 128 GB DDR4 memory, a Hygon Dhyana C86 7185 32-core CPU, and four Hygon DCU accelerators. The nodes adopt fat-tree topology and are connected through a 200 Gbps Infiniband HDR network.

4.1. Accuracy and simulation cases

To validate the accuracy of our parallel SRSCD method, we perform three simulation cases with MISA-SLSCD and compare the results with object KMC (OKMC), serial SRSCD, classical CD, and experiments. The first is a simple case of vacancy accumulation in α-Fe. The remaining two cases are Cu precipitation in Fe-Cu alloy, which is often used as a model alloy for reactor pressure vessels (RPV). In these cases, a simulation box of 32,000 VEs (10 nm length per VE) and periodic boundary

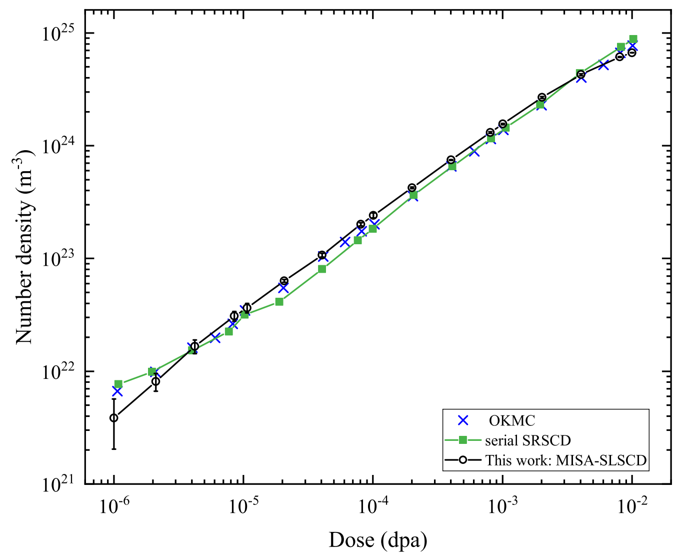


Fig. 6. Evolution of the void number density.

conditions are adopted. Each case is independently simulated at least five times, and we also show the error bars, which represent the standard deviation of the mean values.

4.1.1. Vacancy accumulation in α-Fe

The first case is vacancy accumulation in α-Fe of Dunn et al. [26], which compares the results of OKMC and serial SRSCD. The simulation is carried out at 100 °C, and 20 KeV cascades are implanted in α-Fe with a dose rate of 10^{-4} dpa/s (displacements per atom per second). The diffusion and binding parameters used in the simulation are the same as those in Ref. [26]. The purpose of this case is to compare our MISA-SLSCD and the serial SRSCD to evaluate the accuracy of our parallel method. The void number density of our MISA-SLSCD and serial SRSCD are shown in Fig. 6. We also show the results of OKMC in Ref. [26], which are used to verify the serial SRSCD method. Note that the results of MISA-SLSCD are in good agreement with those of OKMC and SRSCD for the range of dpa shown. Considering that spatial resolution is a feature of the SRSCD method, it is necessary to show the spatial distribution of voids. Fig. 7(b) shows the spatial distribution of void number density under different dpa. For comparison, we also reproduce the spatial distribution results of the serial SRSCD in Fig. 7(a). To quantitatively analyze the difference between our MISA-SLSCD and serial SRSCD, we also show their void number density in each volume element in Fig. 7(c) by using scatter plots. As seen from the visualization results and scatter plots in Fig. 7, our MISA-SLSCD has excellent accuracy.

4.1.2. Cu precipitation in Fe-1.34 at.%Cu under electron irradiation

The second simulation case is Cu precipitation in a Fe-1.34 at.%Cu alloy under electron irradiation. It has been studied by Christien et

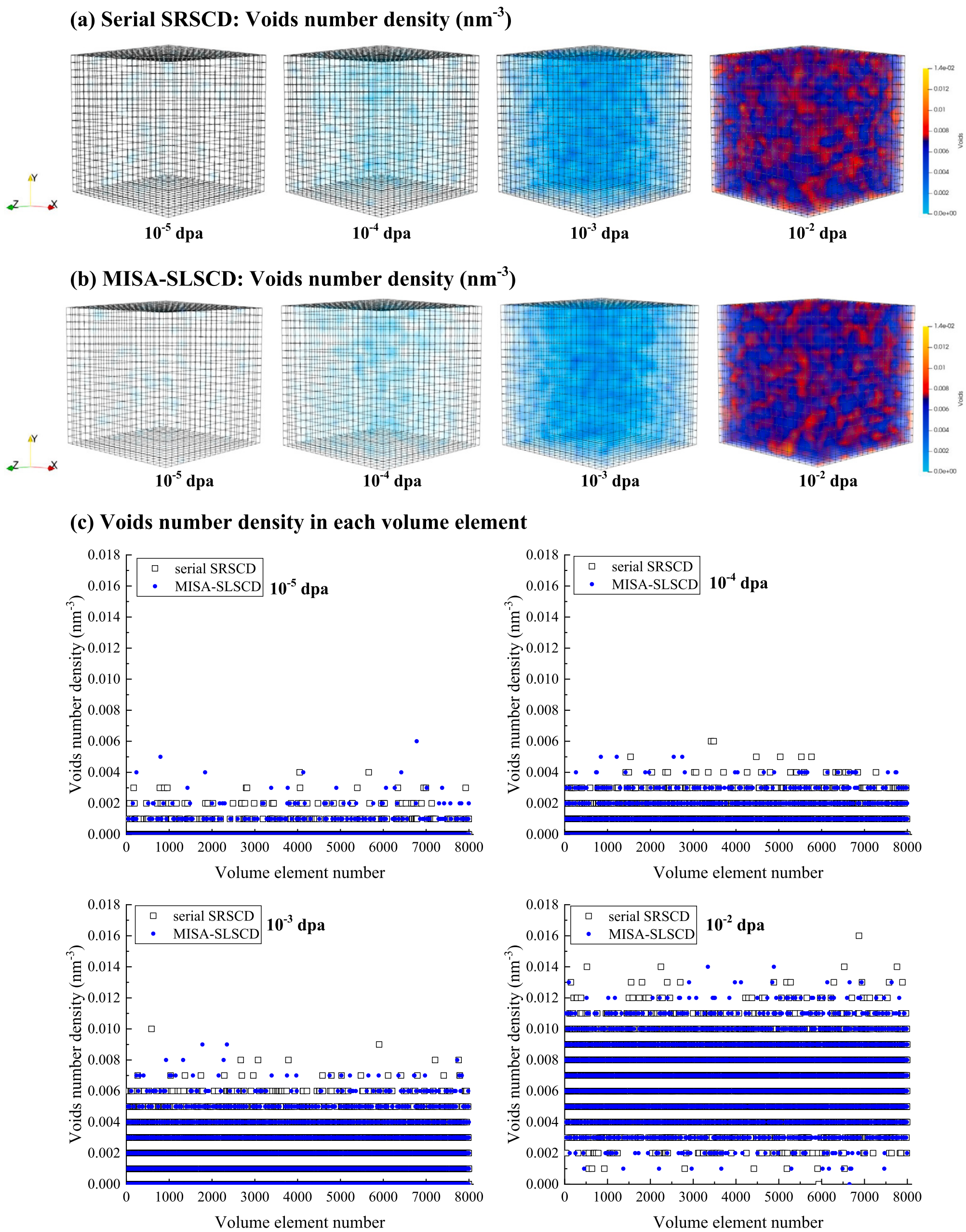


Fig. 7. Visualization and scatter plots of spatial distribution of void number density under different dpa. The number density is the ratio of the number of voids in a VE to the volume of that VE.

al. [50] with the CD method and compared with experiments. The simulation is carried out at 290 °C with a dose rate of 2×10^{-9} dpa/s, and the dislocation density is 10^{-6} nm^{-2}, using the same parameters as those in Ref. [50]. Moreover, this work considers that only vacancies, self-interstitial atoms, and Cu atoms are mobile. Fig. 8 shows the dose evolution of the total number density and mean radius of Cu clusters. The blue line is the result of Christien's CD simulation, and the circle is Mathon's experimental data measured by small-angle neutron scattering (SANS) [51]. It should be noted that only clusters containing more than 10 Cu atoms are used to calculate the number density and mean radius, consistent with Christien et al. [50].

As Fig. 8 shows, initially, the total number density of Cu clusters increases rapidly with dose (as shown in Fig. 8(a)), and the mean radius of Cu clusters increases slowly (see Fig. 8(b)). This result reflects the nucleation and growth of clusters by consuming Cu atoms. Then, the evolution of Cu clusters enters the coarsening stage, with the number density decreasing but the mean radius increasing rapidly. Overall, **our results are not only in good agreement with those of CD but also closer to the experiments**.

4.1.3. Cu precipitation in Fe-0.3 at.%Cu under neutron irradiation

The last simulation case is Cu precipitation in a Fe-0.3 at.%Cu alloy under neutron irradiation. The results are compared with the experimental datas measured by Meslin et al. [52]. We carry out the simulation at 300 °C with a dose rate of 1.4×10^{-7} dpa/s, and the dislocation density changes to 5×10^{-5} nm^{-2}, the same as that used in Ref. [52]. The other simulation parameters are taken from Ref. [14]. Similarly, only vacancies, self-interstitial atoms, and Cu atoms are considered mobile. In this case, 20 KeV cascades are implanted, reaching a total dose of 0.2 dpa. The evolution of the total number density and mean radius of Cu clusters containing more than 10 Cu atoms is shown by the red

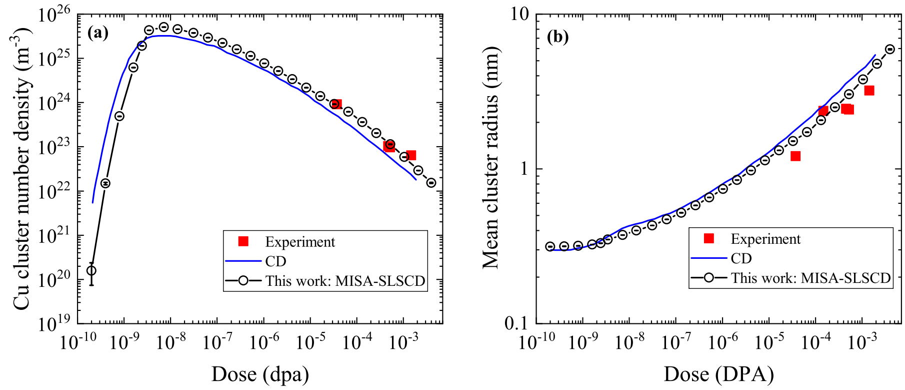


Fig. 8. Precipitation of Cu clusters in a Fe-1.34 at.%Cu at 290 °C under electron irradiation. Evolution of the (a) total number density and (b) mean radius of Cu clusters.

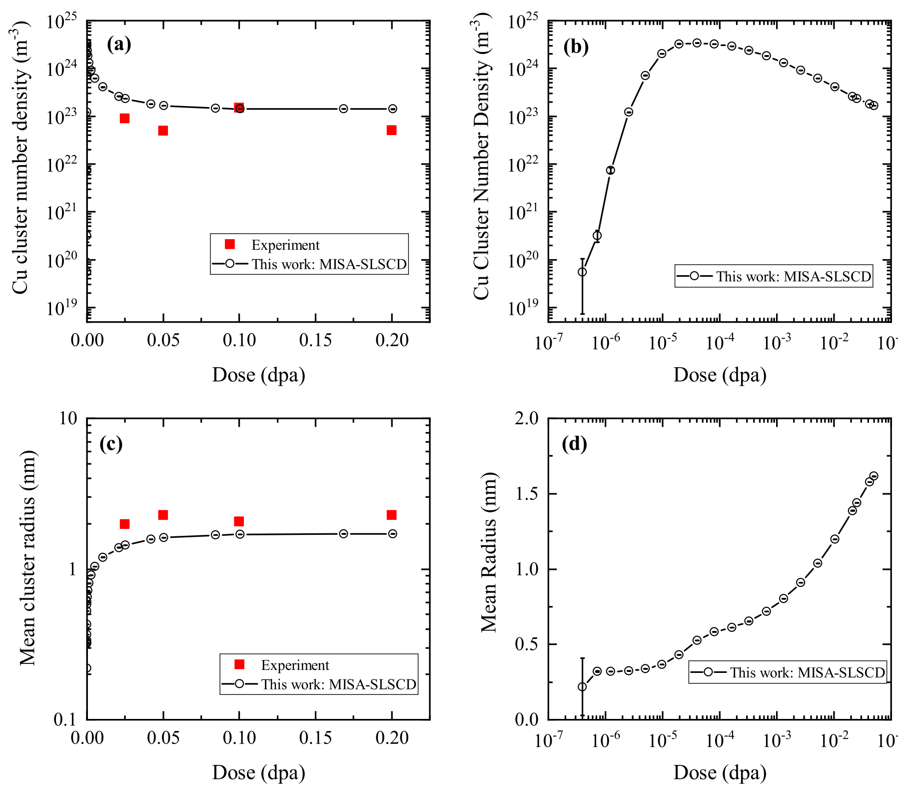


Fig. 9. Precipitation of Cu clusters in a Fe-0.3 at.%Cu at 300 °C under neutron irradiation. Evolution of the (a-b) total number density and (c-d) mean radius of Cu clusters.

lines in Fig. 9. The black circles represent the experimental data measured by Meslin et al. with SANS.

As Fig. 9(a) and Fig. 9(c) show, the number density and mean radius of Cu clusters obtained by our method are very close to those of experiments. It is worth mentioning that 0.1 dpa corresponds to approximately 40 years of operation of pressurized water reactors [52]. We can see from Fig. 9(b) that the coarsening of Cu clusters under neutron irradiation occurs in a higher radiation dose range than that under electron irradiation, and the mean radius of Cu clusters is smaller (Fig. 9(d)). As a case of this process, Fig. 10 visualizes the evolution of different defects simulated by MISA-SLSCD, in which the spatial distribution of Cu clusters, dislocation loops and voids under different dpa are presented. The defects in the system are mainly Cu clusters, and dislocation loops and voids only appear at the later stage of irradiation. This indicates that the precipitation of Cu clusters in metals contributes the most to the radiation damage in RPV.

4.1.4. Accuracy and consistency discussion

One of the main features of the SRSCD method is its ability to provide meaningful statistical variations of the calculation points. So we show the error bars in the five simulation cases to demonstrate stochastic fluctuations in SRSCD simulations. It can be seen that the error bars are obvious only in the early stages of evolution, which corresponds to a low defect number density. And then as the evolution progresses, the number of defects becomes more and more, and the error bar becomes very small. In addition, the mean defect number density obtained by averaging over the series of MISA-SLSCD simulation results (hollow circle) is visibly shifted towards the upper bound of the wide uncertainty intervals. This is due to low defect counts obtained at the beginning stage of evolution, which pushes the lower bounds of the error to rather small values when plotted on a logarithmic scale [25].

As discussed in [53], how to evaluate the accuracy and consistency of simulation results is heavily depended on domain knowledge. In our

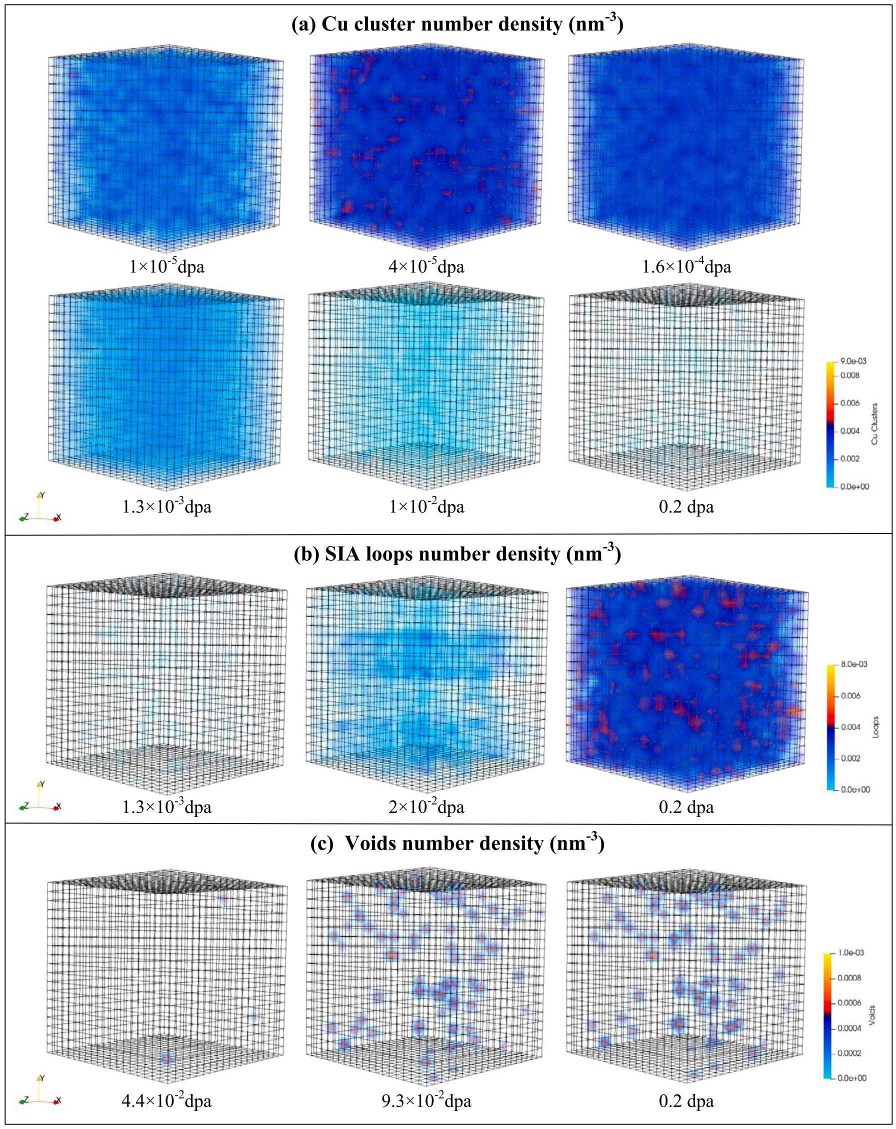


Fig. 10. Visualization of number density distribution of Cu clusters, voids, and dislocation loops under different dpa.

evaluation, we mainly rely on the opinions from domain experts — the end users of our MISA-SLSCD program. Generally, according to the simulation and experimental experience of domain experts, the number density error is within an order of magnitude, and the average radius error dose not exceed ∼30%, which is acceptable in the context of irradiated materials domain. Our domain experts confirm that the simulation results of our program are very accurate and are consistent with their experiential results and other programs, even though as shown in Fig. 6, our result does not perfectly match those of OKMC and serial SRSCD. According to experts' opinions and [51], the results of our program fall within the acceptable error bars of experiment data. Remaining differences between the results found in our program and experiments showed in Fig. 8 and Fig. 9, mainly come from the approximation and simplification of models. For example, the mobility of clusters, the effects of cascade overlap, etc. are ignored. The sources of these differences are therefore considered to be out of the scope of this work. As discussed in 4.1.2, Fig. 8 clearly shows that our results are in better agreement with the experimental results (i.e., the black points) than those of CD. Hence, our conclusions are reasonable and solid.

4.2. Memory usage

To evaluate the memory usage of our DRT structure, we test the memory usage of our MISA-SLSCD and Dunn's SP-SRSCD [30] on 8 CPU cores with the simulation case in subsection 4.1.2. However, it is unfair to compare the memory usage of MISA-SLSCD with the SCD code [25,37] directly through simulation tests. This is because: (1) the SRSCD considers spatial resolution, while the SCD does not, and (2) the SCD code is serial, while SRSCD is parallel.

In the memory usage test, a $8,000$-VE box with a length of 10 nm per VE is used in both MISA-SLSCD and the SP-SRSCD code. Fig. 11 shows the average memory usage per core, and per defect species (containing this species and its associated reactions). As Fig. 11(a) shows, the memory usage increases rapidly and then decreases as the simulation progresses in both MISA-SLSCD and SP-SRSCD code, but at the peak, our MISA-SLSCD achieves an effective memory savings of 39% per core compared to SP-SRSCD. To further analyse the memory savings of our MISA-SLSCD on one defect species, we monitor the average memory usage of one defect species and its associated reactions over simulation time, as shown in Fig. 11(b). The results show that our memory usage remains roughly constant and significantly lower than SP-SRSCD. The

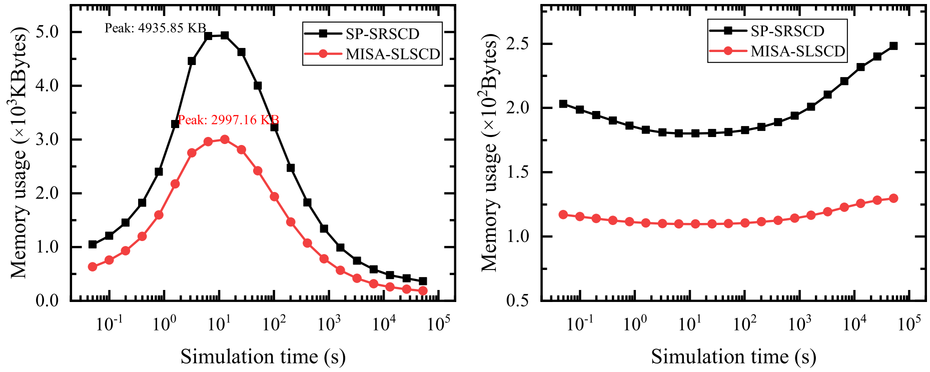


Fig. 11. Memory usage comparison of MISA-SLSCD with the DRT structure and SP-SRSCD. Average memory usage (a) per core and (b) per defect species.

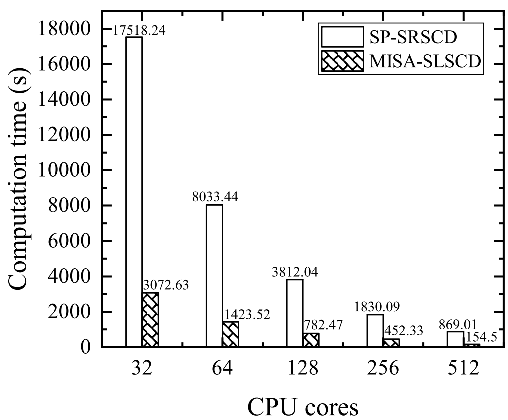


Fig. 12. Computational performance comparison of SP-SRSCD and our MISA-SLSCD.

test results show that our DRT structure can achieve significant memory savings compared to that of SP-SRSCD, which is mainly due to the reduction in redundant storage of partial reactants and products of reactions.

4.3. Performance comparison with SP-SRSCD

To our knowledge, the SP-SRSCD appears to be the only parallel SRSCD procedure at present. Therefore, a comparison between our MISA-SLSCD and SP-SRSCD is very necessary.

4.3.1. Computational performance

In SRSCD simulation, the computation time is mainly spent searching and updating defects and reactions. Therefore, the computational performance is an important evaluation metric for analysing the performance of our parallel SRSCD method. We compare the computational performance of MISA-SLSCD and SP-SRSCD on 32, 64, 128, 256, and 512 CPU cores. All simulations include $48 \times 48 \times 48$ VEs (10 nm length per VE) and a 100 s simulation time. The computation time of both SP-SRSCD and our parallel SRSCD are shown in Fig. 12. It can be seen that MISA-SLSCD saved $75\% \sim 83\%$ of the time costs, compared to that of SP-SRSCD. The improvement is mainly due to the DRL structure and double-grouping search strategy, which speed up the search for defects and reactions.

4.3.2. Parallel performance

In this subsection, we compare the parallel performance of our MISA-SLSCD with that of SP-SRSCD reported in Dunn's work [30]. The model of subsection 4.1.1 is used for the following simulations, and the parallel performance comparison is performed through weak scaling and for two cases: (1) an idealized case in which clustering reaction is not allowed and only Frenkel pair implantation, recombination, and diffusion are allowed; (2) a realistic case in which defect clustering reaction is also allowed. All simulations are performed with 1000 VEs per process and 20 nm length per VE. Simulations with and without clustering reactions all start from 10^{-4} dpa to 10^{-3} dpa, the same as SP-SRSCD [30].

Weak scaling results of the two cases with different dose rates are shown in Fig. 13. The parallel efficiency of MISA-SLSCD is significantly improved under different dose rates compared with that of SP-SRSCD. In the case without clustering, we achieve over 76% parallel efficiency under all dose rates. In the case where clustering reaction is allowed, the parallel efficiencies are still above 72%. This indicates that significant efficiency gains are achieved in MISA-SLSCD, which comes from the combined effect of the adaptive synchronization algorithm and on-demand communication strategy. The former reduces communication frequency, and the latter reduces communication redundancy.

4.4. Scalability

In this subsection, we present the scalability of MISA-SLSCD with a simulation case of Cu precipitation in Fe-0.3 at.%Cu under neutron irradiation with a 100 s simulation time.

We take a fixed simulation volume with 25.6 million VEs (10 nm length per VE) and run the baseline on 1,600 CPU cores for strong scalability. Fig. 14(a) shows the parallel efficiency and total runtime from 1,600 cores to 25,600 cores. As the number of cores increases, the computation time decreases significantly, but communication time gradually becomes dominant. This leads to a decrease in parallelism efficiency. However, with the number of cores varying from 1,600 to 25,600, the parallel efficiency remains above 92%. Fig. 14(b) shows the weak scalability tests. We keep 1000 VEs (10 nm length per VE) per CPU core and initialize the system with 1.6 million VEs on 1,600 cores; this is taken as the baseline. As the number of cores increases from 1,600 to 32,000, the computation time is almost constant, while the communication time increases slightly. The increased communication time is due to the global communication required for calculating the self-correcting time interval τ. We can see that with scaling from 1,600 cores to 32,000 and the total number of VEs increasing from 1.6 million to 32 million, our method achieves over 90% parallel efficiency.

The strong and weak scalability experiments in Fig. 14 demonstrate that MISA-SLSCD is highly scalable and can extend SRSCD simulations to larger scales, facilitating the understanding of damage accumulation behaviours in irradiated materials.

5. Conclusion

The SRSCD method has been developed as a useful method for studying the time evolution of spatially dependent damage in irradiated materials. In this paper, a massively parallel SRSCD method is proposed and implemented. It involves four contributions. First, a dynamic data

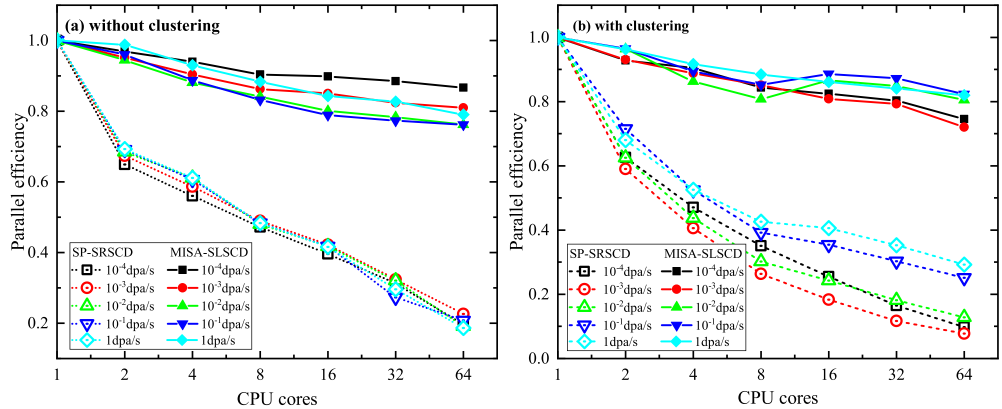


Fig. 13. Weak scalability for different dose rates. Results are compared for cases (a) without and (b) with clustering reactions.

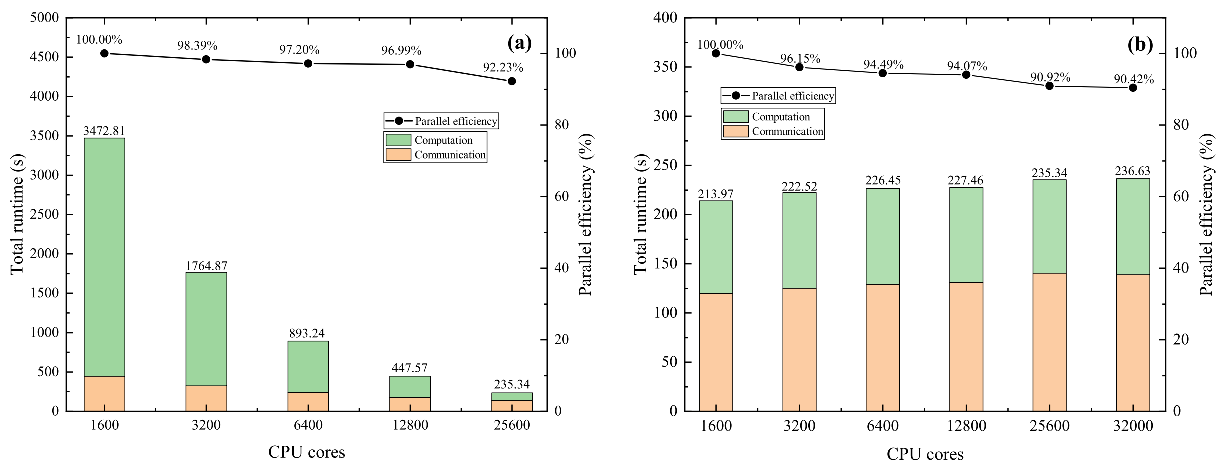


Fig. 14. Strong Scalability for total of 25.6 million VEs (a), and weak scalability for 1000 VEs per core (b). Bars correspond to total runtime of the primary y axis on the left, and the black line corresponds to the parallel efficiency of the secondary y axis on the right. Total runtime and parallel efficiency values are annotated at the top of bars and lines.

structure, Defect-Reaction Tree, is designed to address the rapidly increasing storage and update requirements for defects and associated reactions during SRSCD simulation. The comparative experiment shows that our DRT can achieve significant memory savings. Based on DRL, SRSCD can be easily extended to simulate complex systems with multiple defect species. Then, a DRT-based double-grouping search strategy is proposed to accelerate the selection of reaction events and further improve the computational efficiency. To achieve large-scale parallel SRSCD simulation, we extend the famous synchronous sublattice algorithm to SRSCD, and develop an adaptive synchronization algorithm to balance accuracy and computational efficiency in SRSCD simulations. And an on-demand communication strategy is used to reduce communication redundancy. With these methods, we achieve over 92% strong scaling parallel efficiency on 25,600 CPU cores and over 90% weak scaling parallel efficiency on 32,000 CPU cores. Moreover, comparison with other simulation methods and experimental results shows that our method has a high accuracy for simulating damage accumulation behaviours in irradiation materials. It is worth noting that the proposed methods and optimization strategies in this work can also be effectively applied to other reaction-diffusion simulations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

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